Spring AI is a project by the Spring team that provides integration with **AI/LLM (Large Language Model)** services like OpenAI, Azure OpenAI, Hugging Face, and others. It helps developers easily incorporate generative AI capabilities into **Spring Boot** applications using familiar Spring idioms.

🔍 Key Features of Spring AI

1. **LLM Integration**
   Easily interact with LLMs using OpenAI, Azure OpenAI, Hugging Face, etc.
2. **Prompt Engineering Support**
   Supports templated prompts with variables using Spring's PromptTemplate.
3. **Vector Store Integration**
   Works with vector databases like PostgreSQL (pgvector), Milvus, Pinecone, Redis, etc., for semantic search and retrieval-augmented generation (RAG).
4. **Document Ingestion**
   Ingests data from PDFs, text files, web pages, etc., converts them into chunks, generates embeddings, and stores them in a vector store.
5. **Chat Clients**
   Provides easy interfaces for building chat applications using ChatClient.
6. **Spring Boot Compatibility**
   Seamlessly integrates into Spring Boot 3.x apps with starter dependencies and @Service, @Component support.

🔧 Common Use Cases

- Chatbots and virtual assistants.
- Intelligent Q&A systems over private documents (PDFs, Word, etc.).
- Semantic search.
- Text summarization, translation, and classification.
- Retrieval-Augmented Generation (RAG).

□ Architecture Overview

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[Input Document or Prompt]

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[Prompt Template / Document Loader]

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[Embedding Generator (e.g., OpenAI Embeddings)]

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[Vector Store (e.g., pgvector, Redis, Pinecone)]

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[LLM (OpenAI, Azure, etc.)]

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[Spring AI Output / Chat Response]

Example Maven Dependency

xml

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```
<dependency>
  <groupId>org.springframework.ai</groupId>
  <artifactId>spring-ai-openai-spring-boot-starter</artifactId>
  <version>0.8.1</version>
</dependency>
```

Sample Code: Querying OpenAI

java

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```
@Autowired
private ChatClient chatClient;

public String ask(String question) {
    ChatResponse response = chatClient.call(question);
    return response.getResult().getOutput().getContent();
}
```